

RecovAir: Model-driven airline scheduling tool for disruption recovery

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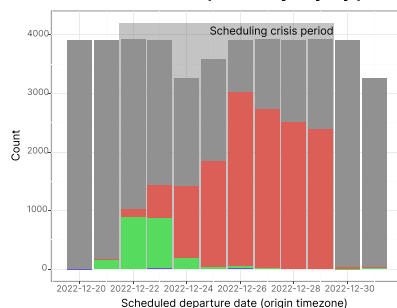
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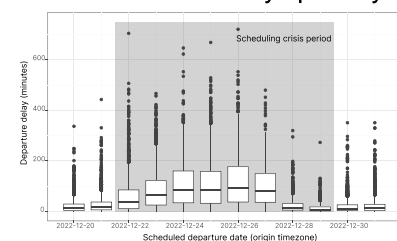
Motivation

- On December 22, 2022, **Winter Storm Elliott** swept over the continental US and crippled the NAS (National Airspace System).
- Most airlines quickly resumed normal operations within 1-3 days, but **Southwest Airlines continued to suffer** from mass cancellations and delays for another week.
- Southwest's initial cancellation package was **over-optimistic**. Unexpected disruptions overwhelmed its schedule and crew assignment solver.

Cancellations per day by type



Distribution of delays per day



~2 million
Passengers stranded

16,700
Flights cancelled

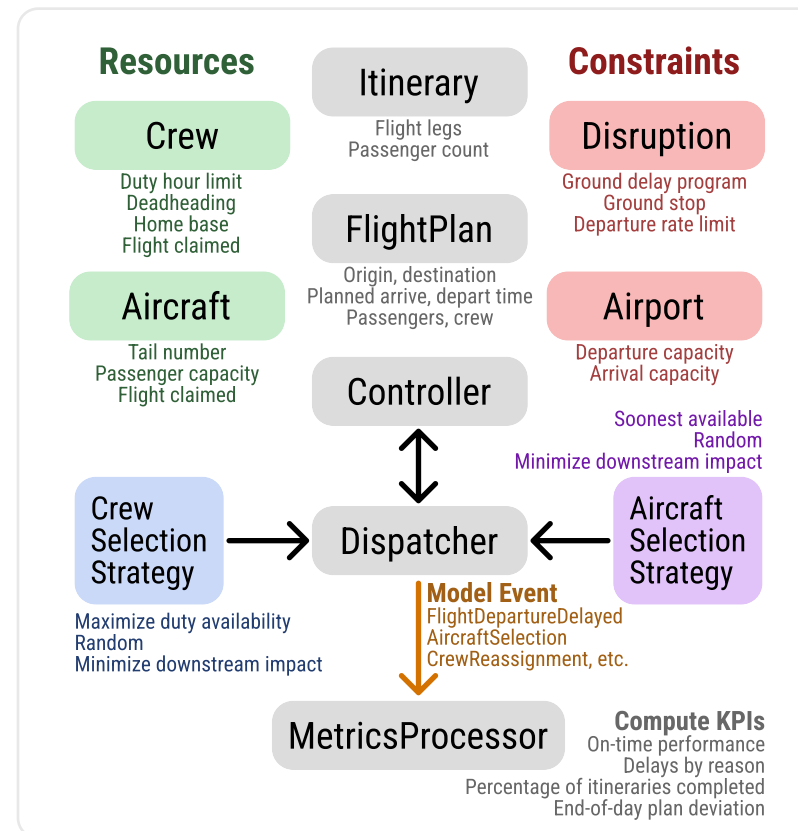
\$820 million
Lost due to crisis

“We had aircraft that were available, but the process of matching those crew members with the aircraft could not be handled by our technology. In our desired state, we have a solver that would be able to do that very quickly and accurately. Our system today cannot do that.”

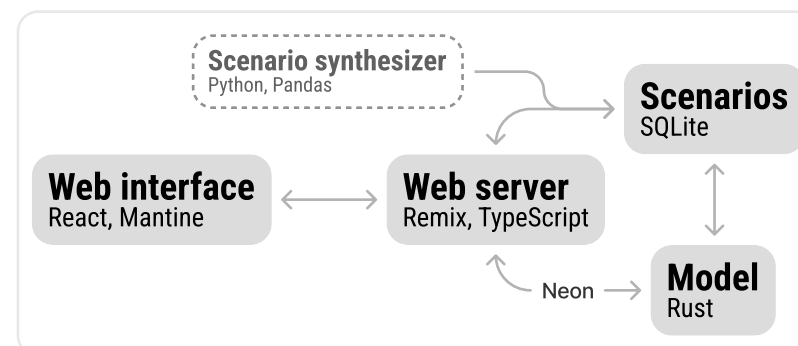
- Andrew Watterson, Southwest Airlines COO, 2023

How might we help airlines quickly develop, evaluate, and deploy schedule recovery tactics in response to unexpected disruptions?

Network model

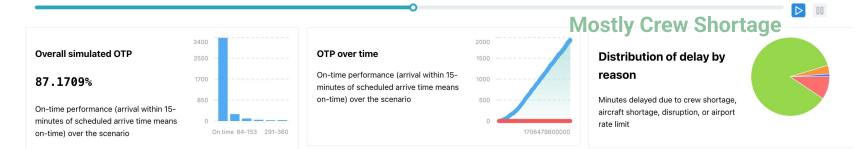
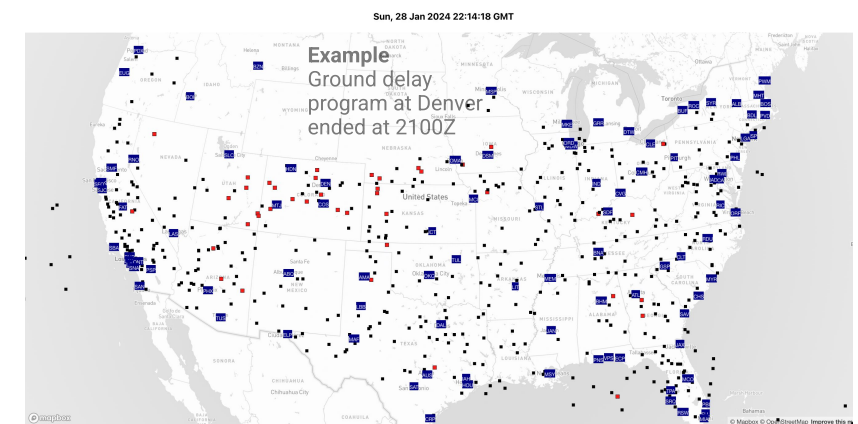


Software architecture



Evaluation

- Scenarios consist of **flight plans** (with assigned aircraft and crew), **airport capacities**, **passenger itineraries**, and **disruption programs**. They are designed to be imported from existing systems at airline network operations centers. Passenger itineraries and crew assignments are **confidential**, so to validate the operation of RecovAir, I developed a tool that **synthesizes plausible scenarios from public flight records** in BTS (Bureau of Transportation Statistics) TranStats datasets.
- Itineraries**: T-100 Passenger Segment + T-100 Passenger Market + Dijkstra's Algorithm for multi-leg origin-destination pairs. **Crew**: Minimum crew needed for schedule * Multiplier



RecovAir successfully simulated Southwest Airlines' flight network from January 28 to January 29, 2024 and demonstrated the **downstream impact of an artificial ground delay program** at Denver.

2802 minutes
Delayed from GDP

87.8% → 88.9%
OTP when GDP removed